SEID 2364 – Week 10 Assignment System Synthesis and Communicating Ethical Analysis

Part 1 – Integrative System Analysis

Selected System: AI Hiring System

The selected system is an AI hiring system used by companies to evaluate and rank job applicants. The system processes resumes, applicant data, and historical hiring patterns to support recruitment decisions.

Regulatory Analysis

Under the EU AI Act, AI hiring systems are generally classified as high-risk systems because they directly influence employment opportunities and economic outcomes. The regulation requires transparency, human oversight, risk management, and data quality standards.

However, some gaps remain. The regulation may classify systems correctly while still failing to address how deployment context changes the severity of harm. For example, the same hiring tool may affect different groups differently depending on company policies and training data.

Bias Pathways and Harm

Bias enters the system mainly through historical training data, ranking models, and human interpretation of AI outputs. If past hiring practices contained discrimination, the AI system may reproduce those inequalities.

The system can create economic harm when applicants lose opportunities unfairly. Social harms may also occur if certain communities become systematically disadvantaged.

Different deployment scenarios create different harms. Internship filtering systems may disadvantage inexperienced students, while promotion systems may reinforce workplace inequality.

EDOCA Responsibility Analysis

Effort is concentrated mainly within technical development and data processing. Applicants invest effort into preparing applications but have little influence over how data is interpreted.

Observability is uneven because developers and companies can inspect system operations more closely than applicants. Applicants often only see the final decision without understanding how it was generated.

Control and authority are mainly held by companies and developers. Recruiters may technically override recommendations, but automation can strongly influence human decisions.

Distortion occurs when complex human qualities are reduced into simplified scores and rankings.

Core Ethical Tension

The core ethical challenge is that the system increases efficiency while simultaneously creating risks of hidden discrimination and accountability gaps. Regulation may classify the system as high-risk, but responsibility remains distributed across developers, recruiters, and companies. This makes it difficult to identify who is responsible when harm occurs.

Part 2 – Paper Framework and Research Organization

Tentative Title

AI Hiring Systems: Bias, Responsibility, and Ethical Challenges in Automated Recruitment

Research Question

How do AI hiring systems distribute responsibility and create ethical risks through bias, limited transparency, and automated decision-making?

Opening Introduction

AI hiring systems are becoming increasingly common in modern recruitment processes. Companies use these systems to reduce hiring time, improve efficiency, and process large numbers of applications. However, automated recruitment systems also create ethical concerns related to discrimination, transparency, and accountability.

This paper examines how AI hiring systems generate harm through biased training data, uneven information access, and distributed responsibility structures. By combining regulatory analysis, bias pathway analysis, and EDOCA responsibility analysis, the paper explores the ethical challenges that emerge when automated systems influence employment opportunities.

Body Section Outline

1. AI Hiring Systems and Automated Decision-Making - Overview of system operation - Role of machine learning in recruitment

2. Bias and Harm Pathways - Historical bias in datasets - Social and economic harms

3. Regulation and the EU AI Act - High-risk classification - Regulatory strengths and limitations

4. Responsibility and Accountability - Distributed responsibility - EDOCA analysis of observability and control

Counterarguments

Some may argue that AI hiring systems improve efficiency and reduce human bias. Others may claim that automation creates more objective decision-making than traditional recruitment methods.

Conclusion Outline

The paper will argue that AI hiring systems can improve efficiency but also create serious ethical risks when bias, transparency issues, and responsibility gaps are ignored. Ethical governance requires both legal regulation and continuous oversight of deployment practices.

Part 3 – Presentation Planning

Presentation Structure

1. Introduction to AI Hiring Systems 2. How Bias Enters the System 3. EU AI Act Classification 4. EDOCA and Responsibility Distribution 5. Ethical Challenges and Harm 6. Conclusion and Recommendations

Main Goal of Presentation

The presentation aims to explain how AI hiring systems operate and why ethical analysis is necessary when automated systems influence employment decisions.

Key Takeaway

AI systems are not neutral technologies. Their ethical impact depends on how they are designed, trained, deployed, and governed within society.